DRUGS · REFLUX

Proton pump inhibitors

Proton pump inhibitors are among the most prescribed drugs in the world and among the most over-prescribed in ENT. They are effective for gastro-oesophageal reflux disease, oesophagitis and peptic ulcer, and for protecting the stomach during NSAID and steroid treatment. For laryngopharyngeal reflux and persistent throat symptoms, however, randomised trials, including the UK TOPPITS trial, have not shown benefit over placebo, so empirical courses should be limited and reviewed.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

PROTON PUMP INHIBITORS

	SUMMARY
Agents	Omeprazole, esomeprazole, lansoprazole, pantoprazole, rabeprazole
How they work	Irreversibly inhibit the H+/K+ ATPase (proton pump) of the gastric parietal cell after activation in acid
ENT uses	Typical reflux symptoms with throat complaints; oesophagitis; after some laryngeal surgery and with laryngeal granulomas; in combination with alginates in suspected laryngopharyngeal reflux (limited evidence); gastroprotection
Adult dose	Omeprazole 20 to 40 mg once daily; lansoprazole 30 mg once daily; twice-daily dosing is often used in suspected laryngopharyngeal reflux, for 8 to 12 weeks, then stopped or reduced
Timing	30 to 60 minutes before breakfast (and before the evening meal if twice daily)

2 CHAPTER TWO Using them in ENT

Practice.

2.1 Throat symptoms

Globus, throat clearing, chronic cough, hoarseness and catarrh have many causes. Examine with nasendoscopy and exclude red flags (dysphagia, weight loss, odynophagia, persistent hoarseness, a neck lump, smoking history) before considering reflux. In the **TOPPITS** trial, lansoprazole 30 mg twice daily for 16 weeks was no better than placebo for persistent throat symptoms. If a trial is given, set a time limit, combine with lifestyle measures and alginates, and stop if there is no benefit.

2.2 Laryngeal granulomas and surgery

Contact granulomas of the vocal process are often treated with anti-reflux therapy, voice therapy and sometimes botulinum toxin; surgery has a high recurrence rate.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Headache, diarrhoea; with long-term use: **C. difficile**, hypomagnesaemia (check magnesium on long-term use, especially with diuretics), fractures, vitamin B12 deficiency, pneumonia, subacute cutaneous lupus, interstitial nephritis; rebound acid hypersecretion on stopping.

3.2 Interactions

Omeprazole and esomeprazole reduce the activation of **clopidogrel** (choose pantoprazole); PPIs reduce absorption of itraconazole capsules, ketoconazole and some tyrosine kinase inhibitors; raise methotrexate levels at high doses.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- The weak evidence in laryngopharyngeal reflux and the **TOPPITS** trial.
- Correct timing of doses.
- Long-term risks and the clopidogrel interaction.

4.2 Common pitfalls

COMMON PITFALLS

- Lifelong PPIs for globus without review.
- Missing a cancer by labelling symptoms as reflux.

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